

AMENDMENT NO. 4 JANUARY 2005
TO
IS 779 : 1994 WATER METERS (DOMESTIC TYPE) —
SPECIFICATION
(Sixth Revision)

(Page 1, clause 2) — Substitute the following for the existing:

‘The Indian Standards listed in Annex A contain provisions which, through reference in this text, constitute provision of this standard. At the time of publication the editions indicated were valid. All standards are subject to revision and parties to agreements based on this standard are encouraged to investigate the possibility of applying the most recent editions of the standards indicated in Annex A.’

(Page 7, clause 12.5) — Add following new clause after 12.5 and renumber the subsequent clauses:

13 SELECTION, INSTALLATION AND MAINTENANCE

Water meters are to be installed and maintained properly, so that they may give desired duty in service. For this purpose, recommendations given in Annex D may be followed.

(Page 7, clause 13.1) — Insert the following new clause at the end:

13.2 Each water meter shall be supplied with a check list giving following:

- a) Check that seal and serial No. of the water meter is intact,
- b) Check test certificate giving test results for pressure tightness, loss of pressure, metering accuracy and minimum starting flow,
- c) Check that before installing meter the line is thoroughly flushed,
- d) Check that the meter is installed according to the direction of flow marked on the meter,
- e) Check that inferential meter is placed horizontally with dial upwards,
- f) Check that the strainer of the meter is not removed, and
- g) Check that the meter has been installed as recommended in the installation drawing, supplied with the check list.

Price Group 2

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ANNEX D
(Clause 13)

**SELECTION, INSTALLATION AND MAINTENANCE OF DOMESTIC
WATER METERS**

Water meters being available in many designs with different end connections and varying flow characteristics, due consideration will have to be given when selecting any one type of meter for a particular situation. Besides the proper selection they have to be properly installed so that they may give desired duty in service without involving frequent repairs and costly maintenance.

D-1 SELECTION

D-1.1 Water meters shall be selected according to flow to be measured and not necessarily to suit a certain size of main. The following points shall govern the selection of meters:

- a) The flow rate shall not exceed $Q_{\max}$, and
- b) The minimum flow to be measured shall be within minimum starting flows.

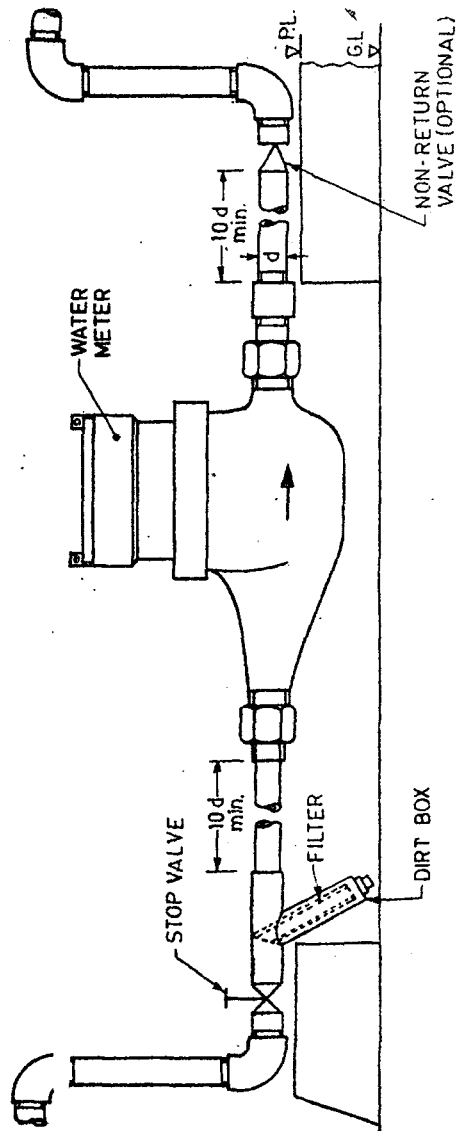
D-1.2 Special care is necessary in selecting the most suitable meter where large rates of flow may exist for short periods, as high rates of flow over short period may cause excessive wear if meter chosen is too small for the duty.

D-1.3 Owing to the fine clearances in the working parts of meters, they are not suitable for measuring water containing sand or similar foreign matter, and in such cases a filter or dirt box of adequate effective area shall be fitted on the upstream side of the meter (*see* Fig. 1). It should be noted that the normal strainer fitted inside a meter is not a filter and does not prevent entry of small particles, such as sand.

D-2 INSTALLATION

D-2.1 A meter shall not be run with free discharge to atmosphere, if the static pressure on the main exceeds 10 meter head of water, otherwise the meter is liable to be over loaded and damaged. For hose connections and similar applications, there shall always be some resistance on the downstream side of the meter.

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d = nominal size of water meter.

FIG. 1 POSITIONING OF WATER METER

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D-2.2 A meter shall be located where it is not liable to get severe shock of water hammer, which might break the piston or damage the rotor, and the position shall be such that it is always full of water, a recommended method of making connection to achieve the purpose is shown in Fig. 1. If the meter body or adjacent pipe becomes partially drained of water, the accumulated air, when passed through the meter, is registered as water, and may cause inaccuracies and perhaps damage. The inaccuracies may be more pronounced in the case of inferential meters. In such situations suitable devices like air release valve may be fitted on the upstream side of the meter. In the case of intermittent water supply system, where there are frequent chances of air locks, the piston of the semi-positive meter often breaks. In such a case, it is advisable to ensure that the top of the meter is below the level of the communication pipe.

D-2.3 Semi-positive meters may be fixed in any position with the dial facing upwards or sideways, and they may be installed in horizontal or vertical pipe runs without affecting wearing properties of accuracy at normal service flows. Where backward flows are anticipated, reflux valves are recommended to be provided. A stop valve should be provided on the upstream side as shown in Fig. 1 to isolate the meter whenever necessary.

D-2.4 Inferential meters shall be installed in position for which they are designed. They shall be placed horizontally with dial facing upwards. However, where meters are to be installed in vertical pipe lines, details shall be as agreed to between the manufacturer and the purchaser.

D-2.5 Turbulent flow of water affects the accuracy of the meter. There shall, therefore, be straight lengths of pipes upstream and downstream of meter for an equivalent length of ten times the nominal size of the meter.

D-2.6 Meters liable to damage by frost shall be suitably protected.

D-2.7 Before installing a meter, the section of the line to be metered shall be thoroughly flushed to remove all foreign matter and when starting up, control valves shall be opened slowly until the line is full, as a sudden discharge may damage the meter.

D-2.8 Water meters may be installed underground, either in the carriage way outside the premises or at a convenient place within the premises. In order to enable the meters to be accessible for periodical reading, inspection, testing and repairs, they shall be housed in suitable water meter boxes. Top of the meter box shall be placed at a slightly higher level than the surrounding ground level so as to prevent ground water entering in and flooding the chamber during rains.